

HiChunk: Evaluating and Enhancing Retrieval Augmented Generation with Hierarchical Chunking

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Outline

- 1 Introduction
- 2 HiCBench
- 3 HiChunk
- 4 Experiments

Chunking in RAG

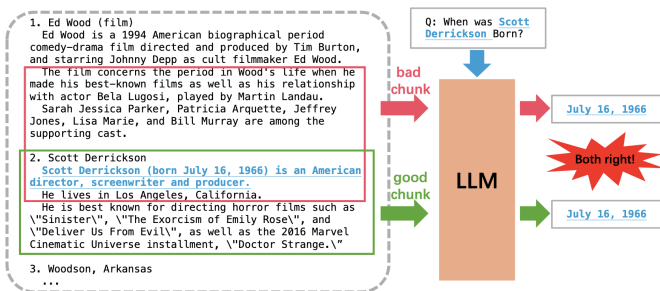
- RAG pipeline:



- The document chunking determines the basic retrieval unit in a RAG system.
 - 1. When chunks are too fine, a complete semantic segment may be broken apart, resulting in insufficient evidence for retrieval recall.
 - 2. When chunks are too coarse, a single chunk contains multiple topics, diluting semantic coherence and filling context with irrelevant text, making it difficult for the retriever to accurately match queries.

Current RAG Benchmark

- Existing RAG systems primarily focus on optimizing the retriever and generator, lacking an independent and accurate method for evaluating chunking. Moreover, current benchmark datasets cannot effectively assess the performance of chunking methods.
- Existing RAG benchmarks face the issue of "evidence sparsity".



- Built on OHRBench; Filter documents with fewer than 4000 words and those exceeding 50 pages; Manually annotate the hierarchical structure.
- Task Definition:
 - Evidence-Sparse QA (T0): The evidence related to the QA is confined to one or two sentences within the document.
 - Single-Chunk Evidence-Dense QA (T1): Evidence sentences related to the QA constitute a substantial portion of the context within a single complete semantic chunk.
 - Multi-Chunk Evidence-Dense QA (T2): Evidence sentences related to the QA are distributed across multiple complete semantic chunks, covering a significant portion of the context.

HiCBench QA Construction

- Using DeepSeek-R1-0528 to generate candidate QA pairs.
- QA Construction:
 - 1.Document Hierarchical Annotation and Summarization: $S \leftarrow LLM_s(D)$.

```
```json
{
 "SECTION 1": "description of section 1",
 "SECTION 1.1": "description of section 1.1",
 ...
 "SECTION n.m": "description of section n.m"
}
```
```

- 2.Generation of Questions and Answers: $(Q, A) \leftarrow LLM_{qa}(S, C)$.
- 3.Ensuring Evidence Completeness and Density: $E \leftarrow LLM_{er}(C, Q, A)$
 $\frac{\text{len}(E)}{\text{len}(C)} \geq 10\%$.
- 4.Ensuring Fact Consistency:

$$F \leftarrow LLM_{fe}(Q, A)$$

$$R' \leftarrow LLM_r(Q, C)$$

$$\text{FactCov} \leftarrow LLM_{fc}(F, R')$$

- The aim is for the fine-tuned LLMs to comprehend the hierarchical relationships within a document and ultimately organize the document into a hierarchical structure.
- identification of chunking points and determination of hierarchy levels.
- Split document D into a list of sentences $S = [s_1, s_2, \dots, s_N]$.
- Chunk point: $(line_number, level)$.

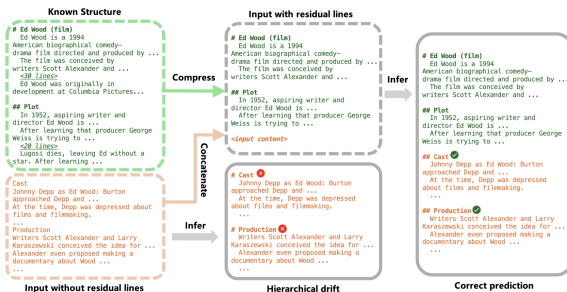
Listing A4: Prompt for model training.

```
You are an assistant good at reading and formatting documents, and you are also skilled at distinguishing the semantic and logical relationships of sentences between document context. The following is a text that has already been divided into sentences. Each line is formatted as: "{line number} @ {sentence content}". You need to segment this text based on semantics and format. There are multiple levels of granularity for segmentation, the higher level number means the finer granularity of the segmentation. Please ensure that each Level One segment is semantically complete after segmentation. A Level One segment may contain multiple Level Two segments, and so on. Please incrementally output the starting line numbers of each level of segments, and determine the level of the segment, as well as whether the content of the sentence at the starting line number can be used as the title of the segment. Finally, output a list format result, where each element is in the format of: "{line number}, {segment level}, {be a title?}".
```

```
>>> Input text:
```

HiChunk iterative inference

- For documents exceeding the model's input length limit L .
- Iterative inference suffers from hierarchical drift phenomenon.



HiChunk iterative inference

- Residual text lines.

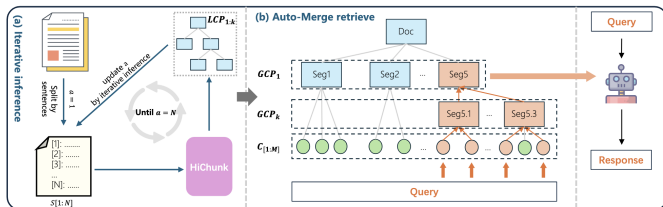
Algorithm 1: iterative inference

input : Document D , Input length L

output : Global chunk points $GCP_{1:k}$

```
1  $S[1 : N] \leftarrow \text{SentTokenize}(D)$ ;  
2  $a \leftarrow 1$ ;  
3  $b \leftarrow \text{argmax}_{\hat{b}}(S[a : \hat{b}] \leq L)$ ;  
4  $\text{res\_lines} \leftarrow \text{None}$ ;  
5  $GCP_{1:k} \leftarrow [] * k$ ;  
6 while  $1 \leq a < b \leq N$  do  
7    $LCP_{1:k} \leftarrow \text{HiChunk}(S[a : b], \text{res\_lines})$ ;  
8    $GCP_{1:k} \leftarrow \text{Merge}(GCP_{1:k}, LCP_{1:k})$ ;  
9   if  $\text{len}(LCP_1) \geq 2$  then  
10     $a \leftarrow LCP_1[-1]$ ;  
11     $\text{res\_lines} \leftarrow \text{None}$ ;  
12   else  
13     $a \leftarrow b$ ;  
14     $\text{res\_lines} \leftarrow \text{ResLines}(GCP_{1:k})$ ;  
15     $b \leftarrow \text{argmax}_{\hat{b}}(S[a : \hat{b}] \leq L)$ ;  
16 return  $GCP_{1:k}$ 
```

HiChunk Auto-Merge Retrieval



- To balance the semantic richness and completeness of recalled contexts, we propose Auto-Merge retrieval algorithm.
- Conditions:

$$\sum_{n \in \mathcal{N} \cap \text{children}(p)} \text{len}(n) \geq \theta^* \text{len}(p)$$

$$\theta^* = \frac{1}{3} \left(1 + \frac{T_{\text{used}}}{T_{\text{max}}} \right)$$

Chunk Accuracy

| Chunk Method | | SC | LC | HC |
|--------------|----------------|--------|--------|---------------|
| Qasper | $F1_{L_1}$ | 0.0759 | 0.5481 | 0.6742 |
| | $F1_{L_2}$ | - | - | 0.5169 |
| | $F1_{L_{all}}$ | 0.1007 | 0.6657 | 0.9441 |
| Gov-Report | $F1_{L_1}$ | 0.0298 | 0.1795 | 0.9505 |
| | $F1_{L_2}$ | - | - | 0.8895 |
| | $F1_{L_{all}}$ | 0.0616 | 0.5631 | 0.9882 |
| HiCBench | $F1_{L_1}$ | 0.0487 | 0.2849 | 0.4841 |
| | $F1_{L_2}$ | - | - | 0.3140 |
| | $F1_{L_{all}}$ | 0.1507 | 0.4858 | 0.5450 |

| Chunk Method | LongBench Score | Qasper | | GutenQA | | OHRBench(T_0) | | HiCBench(T_1) | | | HiCBench(T_2) | | |
|--------------------|-----------------|--------------|--------------|--------------|--------------|-------------------|--------------|-------------------|--------------|--------------|-------------------|--------------|--------------|
| | | ERec | F1 | ERec | Rouge | ERec | Rouge | ERec | FC | Rouge | ERec | FC | Rouge |
| Llama3.1-8B | | | | | | | | | | | | | |
| FC200 | 42.49 | 84.08 | 47.26 | 64.43 | 30.03 | 67.03 | 51.01 | 74.84 | 47.82 | 28.43 | 74.61 | 46.79 | 30.97 |
| SC | 42.12 | 82.08 | 47.47 | 58.30 | 28.58 | 62.65 | 49.10 | 72.14 | 46.80 | 28.43 | 73.49 | 45.28 | 30.92 |
| LC | 42.73 | 87.08 | <u>48.20</u> | 63.67 | <u>30.22</u> | 68.42 | 51.85 | 76.64 | <u>50.84</u> | <u>29.62</u> | 76.12 | <u>49.12</u> | <u>32.01</u> |
| HC200 | 43.17 | 86.16 | 48.09 | <u>65.13</u> | 29.95 | <u>68.25</u> | 51.33 | <u>78.52</u> | 49.87 | 29.38 | <u>78.76</u> | 49.11 | 31.80 |
| +AM | <u>42.90</u> | 87.49 | 48.95 | 65.47 | 30.33 | 67.84 | 51.92 | 81.59 | 55.58 | 30.04 | 80.96 | 53.66 | 33.04 |
| Qwen3-8B | | | | | | | | | | | | | |
| FC200 | 43.95 | 84.32 | 45.10 | 64.50 | 33.47 | 67.07 | 48.18 | 74.06 | 47.35 | 33.83 | 72.95 | 43.45 | 35.27 |
| SC | 43.54 | 82.22 | 44.55 | 58.37 | 32.71 | 62.18 | 46.79 | 71.42 | 46.07 | 33.30 | 72.36 | 42.97 | 34.76 |
| LC | 44.83 | 87.43 | 46.05 | 63.67 | 33.87 | 68.79 | <u>49.28</u> | 75.53 | <u>48.27</u> | 34.12 | 75.14 | <u>46.80</u> | 35.93 |
| HC200 | 43.90 | 86.49 | <u>45.95</u> | <u>65.20</u> | <u>33.89</u> | <u>68.57</u> | 49.06 | <u>77.68</u> | 47.37 | <u>34.30</u> | <u>78.10</u> | 46.20 | <u>36.32</u> |
| +AM | <u>44.41</u> | 87.85 | 45.82 | 65.53 | 34.15 | 68.31 | 49.61 | 81.03 | 50.75 | 35.26 | 80.65 | 49.02 | 37.28 |
| Qwen3-32B | | | | | | | | | | | | | |
| FC200 | 46.33 | 84.32 | 46.49 | 64.50 | <u>44.86</u> | 67.07 | 46.89 | 74.06 | 63.20 | 35.70 | 72.95 | 60.87 | 37.17 |
| SC | 46.29 | 82.22 | 46.39 | 58.37 | 43.59 | 62.18 | 45.43 | 71.26 | 61.09 | 35.64 | 72.36 | 59.23 | 37.09 |
| LC | 47.43 | 87.43 | 46.82 | 63.67 | 44.45 | 68.79 | 47.92 | 75.53 | <u>64.76</u> | 36.15 | 75.14 | <u>62.75</u> | 38.02 |
| HC200 | 46.71 | 86.49 | 46.99 | <u>65.20</u> | 44.83 | <u>68.57</u> | 47.71 | <u>77.68</u> | 63.93 | <u>36.55</u> | <u>78.10</u> | 62.51 | <u>38.26</u> |
| +AM | 46.92 | 87.85 | 47.25 | 65.53 | 44.94 | 68.31 | <u>47.89</u> | 81.03 | 68.12 | 37.29 | 80.65 | 66.36 | 39.37 |

Ablation study

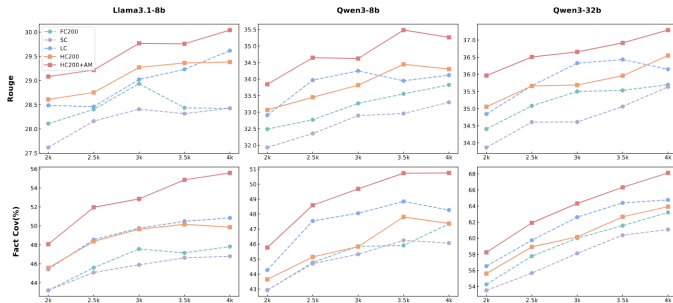


Figure 4: Performance of HiCBench(T_1) under different retrieval token budget from 2k to 4k.

